

UNIVERSITY COLLEGE LONDON
EXAMINATION FOR INTERNAL STUDENTS

MODULE CODE : COMP0078
MODULE NAME : COMP0078 – Supervised Learning
ASSESSMENT : Mock Examination 2024/25
LEVEL : Undergraduate (Masters Level)
DURATION : 3 Hours

Answer ALL FIVE (5) questions.

Marks for each part of each question are indicated in square brackets.

Standard calculators are permitted.

1. (Ridge Regression: Primal, Dual, and Kernelization)

Let $S = \{(\mathbf{x}_i, y_i)\}_{i=1}^m$ with $\mathbf{x}_i \in \mathbb{R}^n$ and $y_i \in \mathbb{R}$. Let $X \in \mathbb{R}^{m \times n}$ be the design matrix whose i -th row is $\mathbf{x}_i^\top$, and let $\mathbf{y} = (y_1, \dots, y_m)^\top$.

a. Consider the ridge regression problem

$$\hat{\mathbf{w}}_\lambda = \arg \min_{\mathbf{w} \in \mathbb{R}^n} \left\{ \frac{1}{m} \|X\mathbf{w} - \mathbf{y}\|^2 + \lambda \|\mathbf{w}\|^2 \right\}, \quad \lambda > 0. \quad (1)$$

(i) Derive the closed-form expression for $\hat{\mathbf{w}}_\lambda$ by differentiating the objective and setting the gradient to zero. [3 marks]

(ii) Show that $\hat{\mathbf{w}}_\lambda$ can be written in the form $\hat{\mathbf{w}}_\lambda = \sum_{i=1}^m \alpha_i \mathbf{x}_i$ for some coefficients $\alpha_1, \dots, \alpha_m \in \mathbb{R}$.

Hint: Rearrange the stationarity condition to isolate $\mathbf{w}$ as a linear combination of the data points. [4 marks]

(iii) Using the result of part (ii), show that the dual coefficients satisfy

$$\boldsymbol{\alpha} = (XX^\top + m\lambda I_m)^{-1} \mathbf{y}.$$

State the computational complexity of solving for $\hat{\mathbf{w}}_\lambda$ via the primal form versus the dual form, and explain when each is preferable. [5 marks]

b. Now suppose we replace $\mathbf{x}_i$ with $\phi(\mathbf{x}_i)$ where $\phi : \mathbb{R}^n \rightarrow \mathbb{R}^N$ is a feature map, with N possibly very large or even infinite.

(i) Show that the prediction function $f(\mathbf{x}) = \hat{\mathbf{w}}_\lambda^\top \phi(\mathbf{x})$ can be written entirely in terms of the kernel function $K(\mathbf{x}_i, \mathbf{x}_j) = \langle \phi(\mathbf{x}_i), \phi(\mathbf{x}_j) \rangle$, without explicit reference to ϕ . [3 marks]

(ii) Consider the kernel $K(\mathbf{x}, \mathbf{t}) = (\mathbf{x}^\top \mathbf{t})^2$ for $\mathbf{x}, \mathbf{t} \in \mathbb{R}^n$. Find an explicit feature map $\phi : \mathbb{R}^n \rightarrow \mathbb{R}^{n^2}$ such that $K(\mathbf{x}, \mathbf{t}) = \langle \phi(\mathbf{x}), \phi(\mathbf{t}) \rangle$. Is this feature map unique? Justify briefly. [3 marks]

(iii) Explain why using the Gaussian kernel $K(\mathbf{x}, \mathbf{t}) = \exp(-\beta \|\mathbf{x} - \mathbf{t}\|^2)$ with kernel ridge regression is advantageous compared to explicitly constructing its associated feature map. What is the dimensionality of the associated feature space? [2 marks]

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2. (AdaBoost: Training Error and Generalization)

Let $S = \{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_m, y_m)\} \subset \mathbb{R}^d \times \{-1, 1\}$ be a training set. Recall the AdaBoost algorithm, which maintains a distribution D_t over $\{1, \dots, m\}$ and produces a final classifier $H(\mathbf{x}) = \text{sign}(\sum_{t=1}^T \alpha_t h_t(\mathbf{x}))$.

a. At each round t , AdaBoost selects a weak learner $h_t : \mathbb{R}^d \rightarrow \{-1, 1\}$ minimizing the weighted error $\epsilon_t = \sum_{i=1}^m D_t(i) \mathcal{I}[h_t(\mathbf{x}_i) \neq y_i]$, and sets $\alpha_t = \frac{1}{2} \ln \frac{1-\epsilon_t}{\epsilon_t}$.

(i) Show that for any example i , the distribution after T rounds can be written as

$$D_{T+1}(i) = \frac{\exp(-y_i f_T(\mathbf{x}_i))}{m \prod_{t=1}^T Z_t}$$

where $f_T(\mathbf{x}) = \sum_{t=1}^T \alpha_t h_t(\mathbf{x})$ and Z_t is the normalisation constant at round t . [4 marks]

(ii) Using the above, prove that

$$\frac{1}{m} \sum_{i=1}^m \mathcal{I}[H(\mathbf{x}_i) \neq y_i] \leq \prod_{t=1}^T Z_t.$$

Hint: First show $\mathcal{I}[H(\mathbf{x}_i) \neq y_i] \leq \exp(-y_i f_T(\mathbf{x}_i))$. [3 marks]

(iii) Show that $Z_t = 2\sqrt{\epsilon_t(1-\epsilon_t)}$, and hence, assuming each $\epsilon_t \leq \frac{1}{2} - \gamma$ for some $\gamma > 0$, prove that the training error satisfies

$$\frac{1}{m} \sum_{i=1}^m \mathcal{I}[H(\mathbf{x}_i) \neq y_i] \leq \exp(-2\gamma^2 T).$$

Hint: Use the inequality $\sqrt{1-x} \leq e^{-x/2}$ for $x \in [0, 1]$. [4 marks]

b. The training error bound above decreases exponentially to zero. Yet in practice, AdaBoost often does not overfit, even as $T \rightarrow \infty$.

(i) Explain, using the concept of **margin**, why the generalization performance of AdaBoost can continue to improve even after the training error reaches zero. [2 marks]

(ii) State the margin-based generalization bound and discuss how the VC-dimension of the weak learner class affects this bound. [2 marks]

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3. (Representer Theorem and Regularized Learning)

Consider a general regularized empirical risk minimization problem:

$$\min_{\mathbf{w} \in \mathbb{R}^N} \left\{ \sum_{i=1}^m V(y_i, \langle \mathbf{w}, \phi(\mathbf{x}_i) \rangle) + \lambda \|\mathbf{w}\|^2 \right\} \quad (2)$$

where $V : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ is a loss function differentiable in its second argument, $\phi : \mathbb{R}^n \rightarrow \mathbb{R}^N$ is a feature map, and $\lambda > 0$.

a. Recall the Representer Theorem: state it precisely. [2 marks]

b. Prove the Representer Theorem. Namely, show that if $\mathbf{w}^*$ minimizes (2), then there exist $\alpha_1, \dots, \alpha_m \in \mathbb{R}$ such that

$$\mathbf{w}^* = \sum_{i=1}^m \alpha_i \phi(\mathbf{x}_i).$$

Hint: Differentiate with respect to $\mathbf{w}$ and rearrange. [5 marks]

c. Using the Representer Theorem, show that the objective (2) can be rewritten purely in terms of $\boldsymbol{\alpha} = (\alpha_1, \dots, \alpha_m)^\top$ and the kernel matrix $\mathbf{K}$ where $K_{ij} = K(\mathbf{x}_i, \mathbf{x}_j) = \langle \phi(\mathbf{x}_i), \phi(\mathbf{x}_j) \rangle$. [3 marks]

d. Now specialize to the case $V(y, \hat{y}) = (y - \hat{y})^2$ (squared loss). Derive the closed-form expression for $\boldsymbol{\alpha}^*$ in terms of $\mathbf{K}$, $\mathbf{y}$, and λ . [4 marks]

e. Suppose instead $V(y, \hat{y}) = \max(0, 1 - y\hat{y})$ (hinge loss). Can you still write a closed-form solution for $\boldsymbol{\alpha}^*$? If not, explain what form the optimization takes when written in terms of $\boldsymbol{\alpha}$ and $\mathbf{K}$, and identify the connection to a well-known learning algorithm. [3 marks]

f. Briefly discuss the role of the kernel K in allowing us to work in potentially infinite-dimensional feature spaces without explicitly computing $\phi(\mathbf{x})$. Give one example of a kernel whose associated feature space is infinite-dimensional. [3 marks]

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4. (PAC Learning, VC Dimension, and Generalization)

a. Recall the definitions of:

(i) PAC learnability of a hypothesis class H . [2 marks]

(ii) The VC dimension of H . [2 marks]

b. Let $\mathcal{X} = \mathbb{R}$ and $H = \{x \mapsto \text{sign}(x - \theta) : \theta \in \mathbb{R}\}$ be the class of threshold functions.

(i) Compute $\text{VCdim}(H)$. Provide a full proof: exhibit a set that is shattered and argue that no larger set can be shattered. [3 marks]

c. Let $\mathcal{X} = \mathbb{R}^n$ and $H = \{x \mapsto \text{sign}(\langle \mathbf{w}, \mathbf{x} \rangle) : \mathbf{w} \in \mathbb{R}^n\}$ be the class of homogeneous halfspaces (no bias).

(i) Show that $\text{VCdim}(H) = n$.

Hint: For the lower bound, use the standard basis vectors. For the upper bound, argue using linear dependence of any $n + 1$ vectors in $\mathbb{R}^n$. [5 marks]

d. Let H be a finite hypothesis class with $|H|$ elements.

(i) Prove that H is PAC learnable with sample complexity $m_H(\epsilon, \delta) \leq \frac{\log(|H|/\delta)}{\epsilon}$.

Hint: Define a “bad” hypothesis as one with $L_{\mathcal{D}}(h) > \epsilon$. Use the fact that $L_S(h) = 0$ for a bad hypothesis occurs with probability at most $(1 - \epsilon)^m$, then apply the union bound over all bad hypotheses. [5 marks]

e. In the **agnostic** PAC setting (where the realizability assumption is dropped):

(i) State the agnostic PAC learnability definition and the key difference from the realizable PAC setting. [2 marks]

(ii) Using Hoeffding’s inequality and the union bound, prove that a finite class H is agnostically PAC learnable with sample complexity $m_H(\epsilon, \delta) \leq \frac{2 \log(2|H|/\delta)}{\epsilon^2}$. [4 marks]

(iii) Compare the dependence on ϵ in the realizable versus agnostic cases. Explain intuitively why the agnostic rate is slower. [2 marks]

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5. (Online Learning: Experts and Linear Classifiers)

a. Consider an online learning problem with n experts. At each round $t = 1, \dots, T$:

- The learner assigns a distribution $\mathbf{v}_t \in \Delta_n = \{\mathbf{v} \in [0, 1]^n : \sum_{i=1}^n v_i = 1\}$ over experts.
- Nature reveals a loss vector $\ell_t \in [0, 1]^n$.
- The learner incurs expected loss $\mathbf{v}_t \cdot \ell_t$.

The Hedge algorithm updates $v_{t+1,i} \propto v_{t,i} e^{-\eta \ell_{t,i}}$, with $\mathbf{v}_1 = (1/n, \dots, 1/n)$.

(i) Define the **regret** of the Hedge algorithm with respect to the best single expert. [1 mark]

(ii) Let $d(\mathbf{u}, \mathbf{v}) = \sum_{i=1}^n u_i \ln(u_i/v_i)$ denote the KL divergence. Show that for any $\mathbf{u} \in \Delta_n$:

$$d(\mathbf{u}, \mathbf{v}_t) - d(\mathbf{u}, \mathbf{v}_{t+1}) \geq \eta(\mathbf{v}_t \cdot \ell_t - \mathbf{u} \cdot \ell_t) - \frac{\eta^2}{2} \sum_{i=1}^n v_{t,i} \ell_{t,i}^2.$$

Hint: Use $e^{-x} \leq 1 - x + x^2/2$ and $\ln(1+x) \leq x$. [6 marks]

(iii) By summing over $t = 1, \dots, T$ and choosing $\mathbf{u}$ as the indicator vector for the best expert, derive the regret bound

$$\sum_{t=1}^T \mathbf{v}_t \cdot \ell_t - \min_{i \in [n]} \sum_{t=1}^T \ell_{t,i} \leq \frac{\ln n}{\eta} + \frac{\eta T}{2}.$$

[4 marks]

(iv) Choose η to minimize the above bound. What is the resulting regret bound? State it as a function of n and T . [2 marks]

b. Now consider the Perceptron algorithm for online linear classification. Data arrives as $(\mathbf{x}_t, y_t) \in \mathbb{R}^n \times \{-1, +1\}$ with $\|\mathbf{x}_t\| \leq R$. Assume there exists $\mathbf{v} \in \mathbb{R}^n$ with $\|\mathbf{v}\| = 1$ such that $y_t(\mathbf{v} \cdot \mathbf{x}_t) \geq \gamma > 0$ for all t .

(i) State the Perceptron update rule. [1 mark]

(ii) Prove that the total number of mistakes M satisfies $M \leq \frac{R^2}{\gamma^2}$.

Hint: Establish two facts: (a) if a mistake occurs on trial t , then $\|\mathbf{w}_{t+1}\|^2 \leq \|\mathbf{w}_t\|^2 + R^2$; (b) after M mistakes, $\mathbf{w} \cdot \mathbf{v} \geq M\gamma$. Combine using Cauchy–Schwarz. [6 marks]

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